

The Entity Relationship Diagram models the structure of MoMo SMS; it is organized in 5 entities: Users, Transactions, Transaction_Categories, System_Logs, and User_Categories.

The Users entity will help with storing individuals who participate in different transactions, and are identified uniquely by the user_id and the phone number. The transactions entity records every transaction, including the amount, the fee charged, the balance remaining after the transaction, the status (successful or failed), the date, as well as the receiver_id and the sender_id because every transaction has a receiver and a sender.

The transaction_categories entity classifies the transactions into categories such as bank transfer, airtime purchase, and others. Having a categories table helps with reporting and organizing easily. The system_logs entity tracks data processing events linked to specific transactions; each log entry contains a message describing the transaction and a timestamp recording when it occurred, and the transaction_id foreign key links each log entry to a specific transaction, allowing us to trace each transaction back to its source.

Lastly, the user_categories junction table, which represents the many to many relationships between users and categories. This is to represent that a single user can engage in multiple types of transactions; for example, a user can send money, buy airtime, and other transactions.

The relationships in our ERD include 1:M and M:N, which illustrate that a user can make many transactions over time, with each transaction being initiated by a single sender (User – Transactions 1:M). One category appears in many transactions, but each transaction has only one category (Categories - Transactions 1:M). One transaction can have many logs (Transactions – System_Logs 1:M). A user can use different types of transactions, and each transaction type can be used by many users (Users – Categories M:N).